Introduction

MADRE: Méthodologie pour l'Analyse de Données Réseaux
(Methodology for Network Data Analysis)

Francesca Fossati

2025-2026

Outline

1. Objective and organization
2. ML and AI
3. Network data analysis
4. Basics on statistics and probability

Objective and organization

The cours has a double objective :

1. to present some basic concept on data analysis and to apply them to networking data;
2. to present the methodology for a (data science) project.

MADRE is organized as follow:

- 14 weeks
- 4 hours on monday morning
  - 8h30-10h30 DATA ANALYSIS
  - 10h45-12h45 METHODOLOGY
- Language: English

Contacts

DATA ANALYSIS

- Francesca Fossati (francesca.fossati@lip6.fr) : Cours sessions
- Elif Ohri (elif-ebru.ohri@lip6.fr) : Laboratory session
- Guillaume Fraysse (guillaume.fraysse@orange.com) : Expert session

METHODOLOGY

- Francesca Fossati (francesca.fossati@lip6.fr)
- Marcelo Dias de Amorim(marcelo.amorim@lip6.fr)
- Bin-Minh Bui-Xuan (Binh-Minh.Bui-Xuan@lip6.fr)

Data analysis

	Day	Data analysis	Subject
week 1	15-sept	CM	Introduction
week 2	22-sept	CM	Regression
week 3	29-sept	TP	Regression
week 4	06-oct	CM	Cross validation
week 5	13-oct	CM	Classification
week 6	03-nov	TP	Cross validation
week 7	10-nov	TP	Classification
week 8	17-nov	CM	Clustering
week 9	01-déc	TP	Clustering
week 10	08-déc	CM	PCA / anomaly detection
week 11	15-déc	CM	DNN
week 12	05-janv	TP	PCA / anomaly detection
week 13	12-janv	CM (Expert)	RL
week 14	19-janv	TP	DNN

CM = lecture course TP = technical laboratory

Data analysis

- TPs are on github (link on moodle)
- You can work in a group (2 people max).
- You should fill, before each TP, a google form.
- The TP deadline is the sunday at 23:59.
- You should upload the code on Moodle before the deadline time.

Methodology

Some classes will be dedicated to methodology aspects (e.g. ML pipeline, reading and critical comprehension, writing, presentation,...) and some classes will be dedicated to the project milestone and presentation.

More info in the methodology class.

Final grade

40% EXAM

- Data analysis and methodology questions

40% PROJECT

- Data analysis and methodology aspects

20% TP+HOMEWORKS+WORKS IN CLASS

- Data analysis and methodology

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Definitions

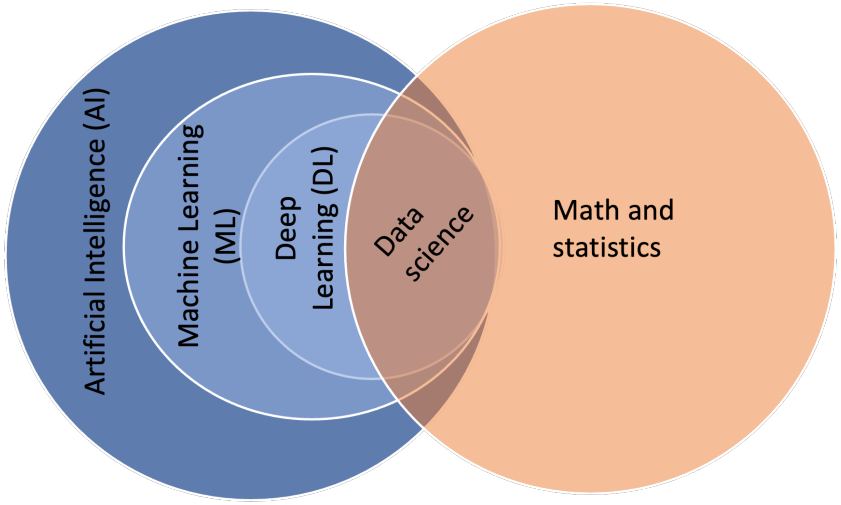
Statistic, Machine Learning (ML) and Artificial Intelligence (AI), Data Science: what is the difference?

Statistics vs ML

Statistics	Machine Learning
Understanding relationships between variables and explaining phenomena from data.	Build high-performance models. The emphasis is on performance, not necessarily on interpretation.
Based on theoretical assumptions (normality, linearity, etc.) and explicit models (regressions, hypothesis testing, etc.).	Less dependent on hypotheses; often empirical; favors complex models (trees, random forests, neural networks).
Focuses on **statistical significance**	Focuses on **generalization ability**
Models are often simple and interpretable (e.g. coefficients in linear regression).	Models can be black boxes (e.g. deep learning), so difficult to interpret, but very powerful.

Definitions

- ML is a subset of AI.
- AI has the goal of simulating human intelligence in machines while ML of developping algorithms that can learn patterns from data and make predictions or decisions.
- AI subfield are: ML, NLP, Computer vision, Robotics, Expert systems.



- Data science has the goal of understand data and help in decision-making. It includes: Data cleaning and preprocessing, Data analysis and visualization, Statistical modeling, Predictive modeling using ML, Communicating findings (dashboards, reports).

Definitions

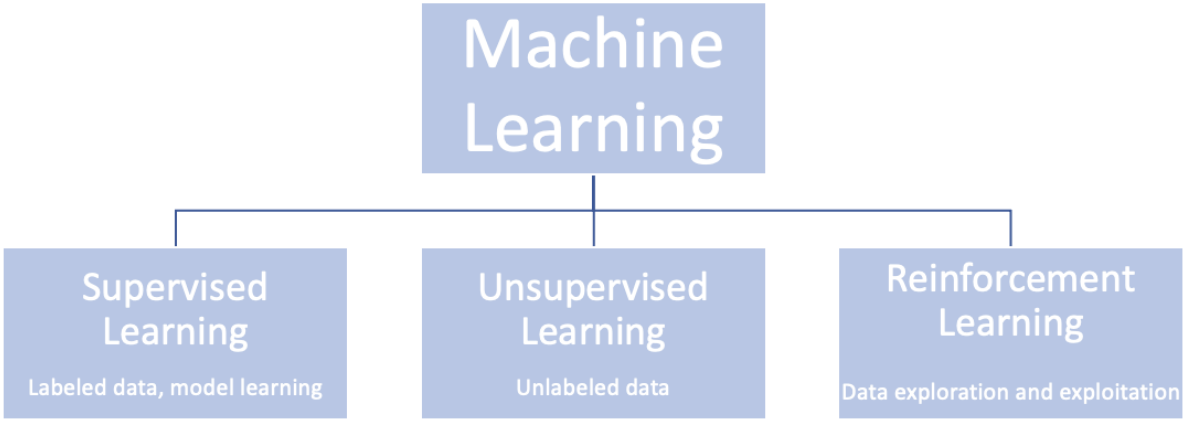
Been around for a very long time...

"Machine Learning, is the field of study that gives computers the ability to learn without being explicitly programmed" (Arthur Samuel, 1959)

...Why now?

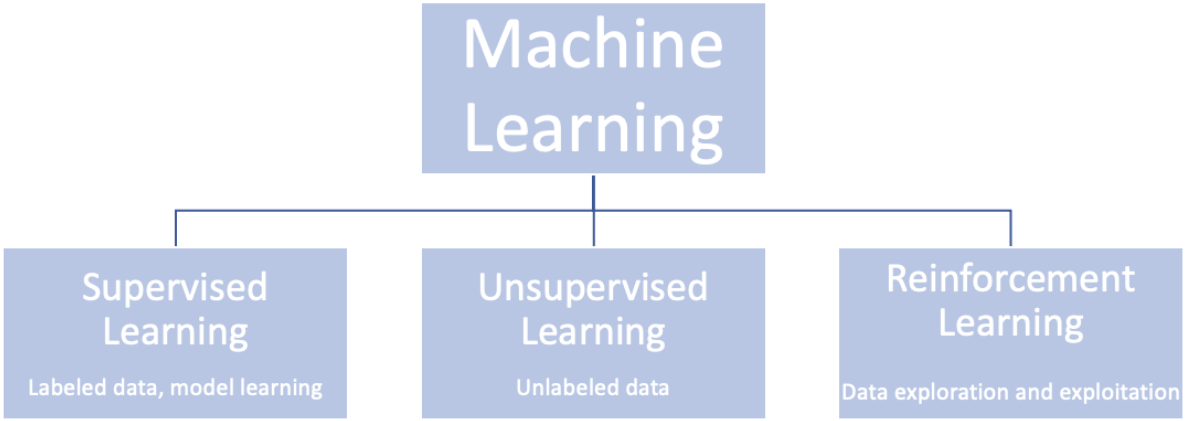
Sufficient and cheap computational power and lots, lots, lots of (labeled) data available e.g. Facebook and Google photos, WWW, ...

Taxonomy of ML methods



+ 2 methods : Semi-supervised Learning and Recommendation systems

Taxonomy of ML methods



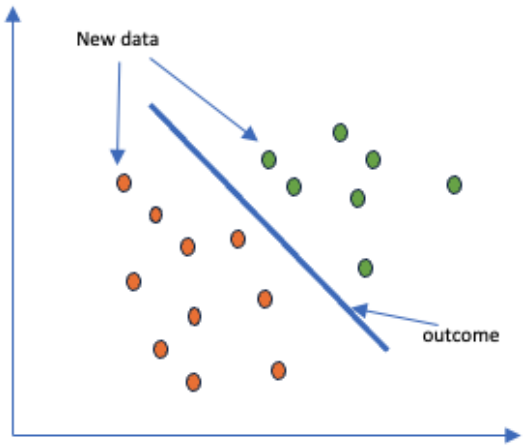
+ 2 methods : Semi-supervised Learning and Recommendation systems

All the methods require a common element: DATA!

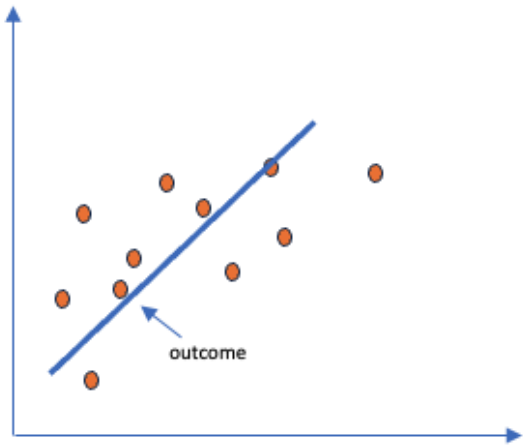
⚠Make a distinction between static and dynamic environments

Supervised Learning

- Classification: image classification, facial recognition, intrusion detection, etc.
- Regression: weather forecasting, market forecasting, etc.



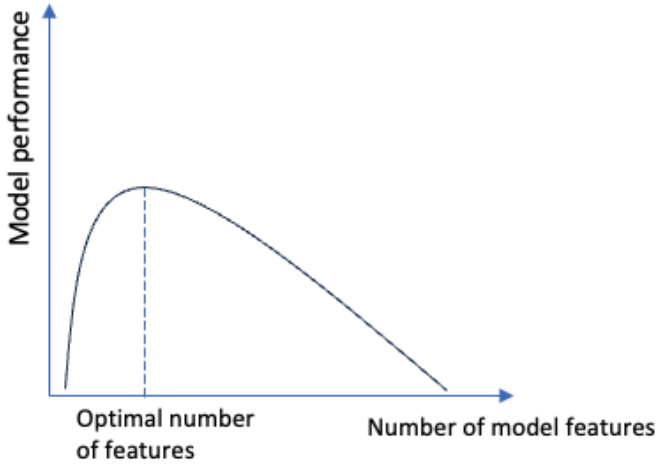
Classification



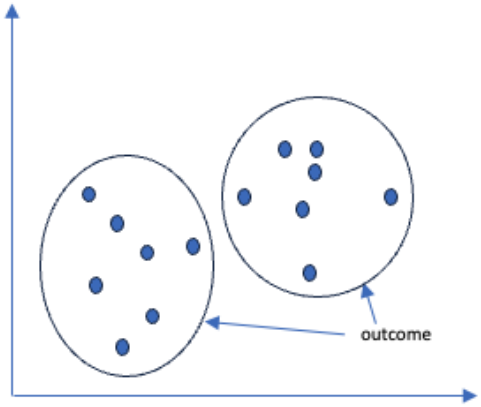
Regression

Unupervised Learning

- Dimensionality reduction: big data visualisation, feature elicitation, etc.
- Clustering: targeted marketing, ...



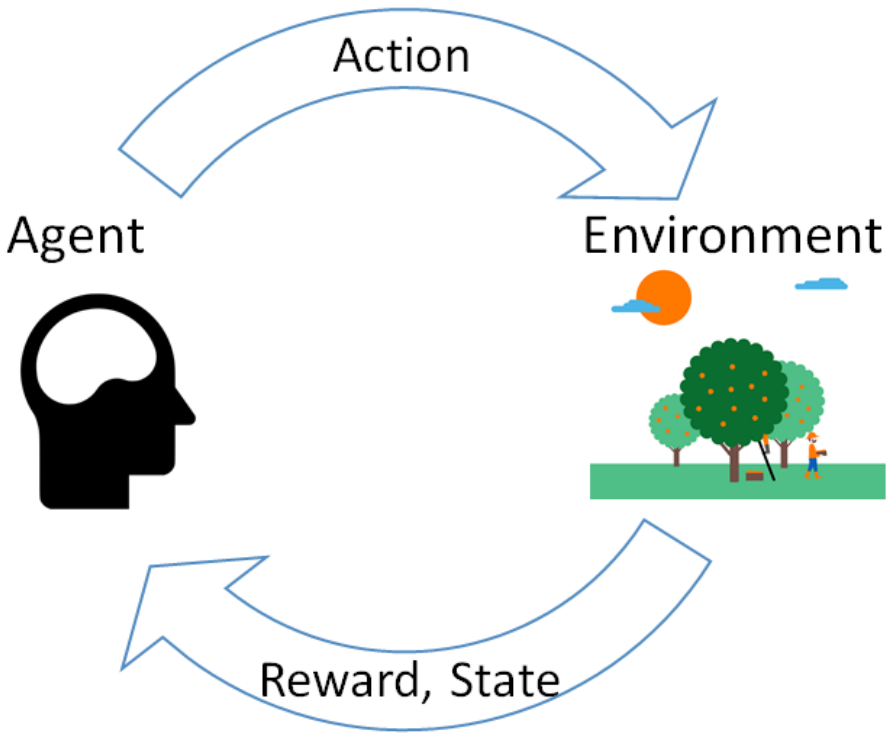
Dimensionality reduction



Clustering

Reinforcement Learning

Real time decision : Autonomous cars, personal assistant (Microsoft Cortana, Apple Siri, ...), etc...



Basic concept

- Train/test "protocol": we work on the training set and test the algorithm on the test set
- Parsimony: increasing the parameters of the model can produce a better solution but if it adapts too much to the dataset can not work good on news data (overfitting)
- Randomness: random decisions allow algorithms to be a bit imprecise (avoid overfitting)

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Examples

Telecom networks offer the infrastructure for ML.

- **Traffic**: prediction, classification, adaptive routing
- **Performance**: congestion control, resource management, QoS/QoE management
- **Anomaly detection**: hardware/software failure
- **Security**: intrusion detection, DoSor DDoS attacks

Traffic

- **Prediction**: Forecast future traffic from previously observed data: Time series forecasting through ARMA models (auto-regressive moving average)
- **Classification**: Associate network traffic to pre-defined classes, e.g. HTTP, FTP, WWW, DNS, P2P or applications, e.g. Skype, YouTube, Netflix...
  Features: port number, packet payload, host behaviour, flow features, QoS requirements.
- **Routing**: Select a path for packet transmission with an objective: cost minimisation, link utilisation, QoS provisioning, etc. Use of Reinforcement Learning techniques, to explore the environment without supervision.

TCP congestion control

TCP protocol limits the packet sending rate when congestion is detected but TCP recognizes and handles all packet losses as network congestion (buffer overfow).
A packet loss can be due to other reasons.
Solution: Classify the cause of packet loss and reduce TCP transmission rate only when congestion.
Also, learn the appropriate window reduction per congestion event!
Features: inter-arrival time, round-trip time, one-way delay.

Security

Attacks can compromise the network's availability and resources

+

cost billions in damage and recovery and may have impact on business reputation.

Current Security measures include : Encryption of network traffic, Anti-viruses, Firewalls, etc.
Extra protection: Intrusion Detection/Prevention monitoring the network for malicious / anomalous activities, and patterns (=attack signatures) in big datasets that deviate from normal behaviour.
Use of unsupervised learning, clustering methods.

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Descriptive statistics

Measures of central tendency or dispersion in data:

- **Mean**

$$\mu = \sum_{i=1}^{N} \frac{X_i}{n}$$

where X_i represents each value in the dataset, and N is the total number of values.

- **Variance**

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^{n} (X_i - \mu)^2$$

where X_i represents each value in the dataset, μ is the mean and N is the total number of values.

- **Standard deviation**

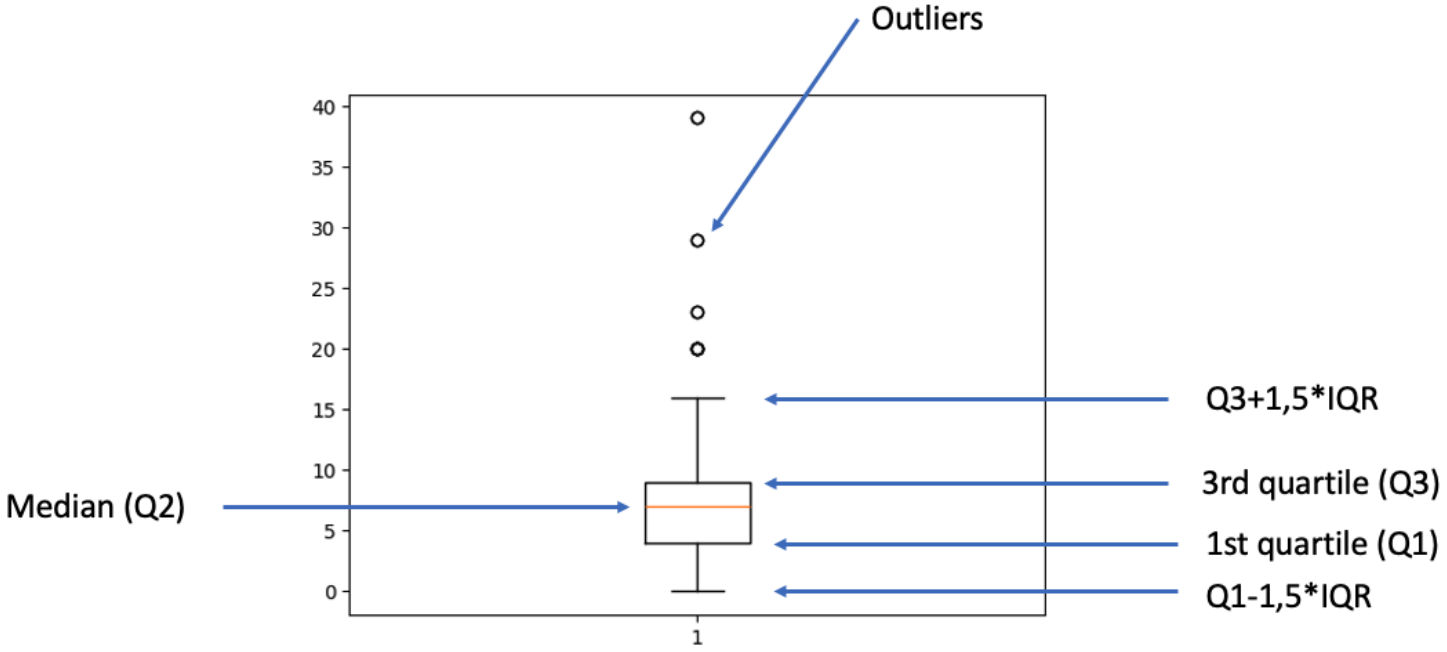
$$\sigma = \sqrt{\sigma^2}$$

Descriptive statistics

- **Percentiles**
  A percentile is a value below which a certain percentage of the data falls. For example, the 90th percentile is the value below which 90 % of the data falls.
- **Median** It is the 50th percentile (middle value).

Boxplot and outliers

A boxplot (also known as a box-and-whisker plot) is a graphical representation of a dataset that shows the **distribution**, **central tendency**, and **spread** of the data.



IQR=inter-quartile range

Correlation

Correlation is a statistical measure that describes the strength and direction of the relationship between two variables.

Pearson Correlation Coefficient

$$r = \frac{\sum(X_i - \overline{X})(Y_i - \overline{Y})}{\sqrt{\sum(X_i - \overline{X})^2 \sum(Y_i - \overline{Y})^2}}$$

where:

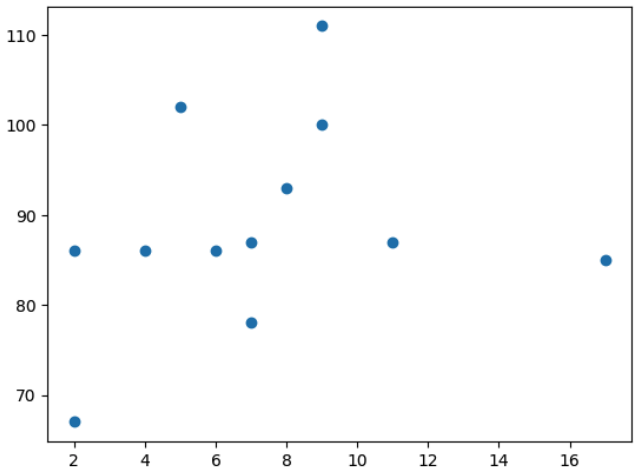
- X_i and Y_i are the individual data points for variables X and Y
- $\overline{X}$ and $\overline{Y}$ are the means of the variables X an

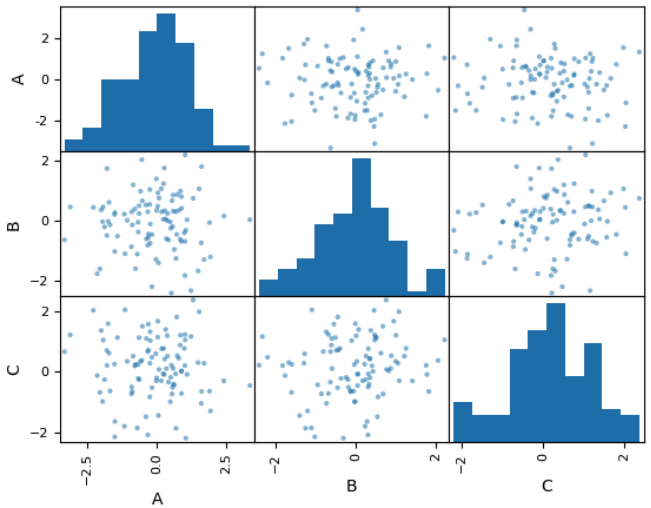
Correlation

- r=1: Perfect positive linear correlation. As one variable increases, the other always increases in a perfectly linear fashion.
- r=-1: Perfect negative linear correlation. As one variable increases, the other always decreases in a perfectly linear fashion.
- r=0: No linear correlation. The variables do not have any linear relationship.
- r between 0 and 1: The closer is to 1, the stronger the positive linear relationship.
- r between -1 and 0: The closer is to -1, the stronger the negative linear relationship.

Correlation

Scatter plots and scatter matrix can help us to visualize the correlation between 2 variables.

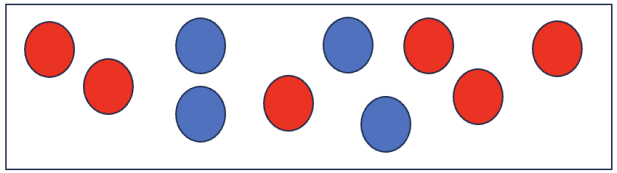




Event probability

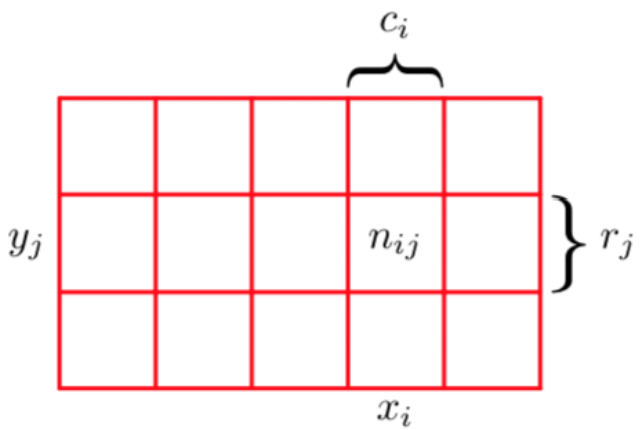
The probability of an event is the fraction of times that the event occurs out of the total number of trials, in the limit that the number of trials goes to infinity

- Each probability must lie in the interval [0; 1].
- If the events are mutually exclusive and if they include all possible outcomes (e.g. red, or blue balls) these probabilities must sum to 1.



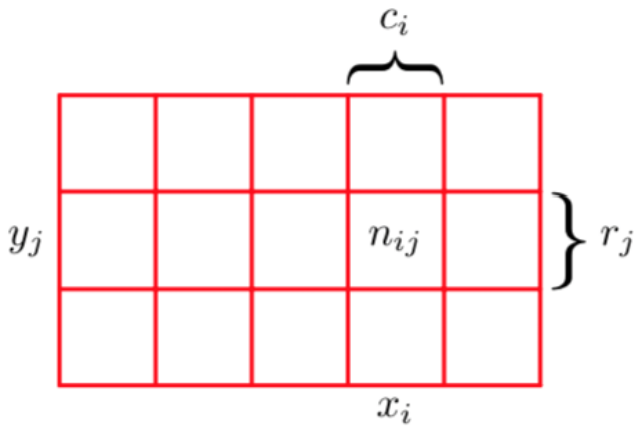
Joint probability

- Two random variables: X and Y
- X takes any value x_i, $i = 1, \ldots, M$
- Y takes any value y_i, $i = 1, \ldots, L$
- We sample both the random variable N timrd each
- The number of time we find $(X = x_i; Y = y_j)$ is $n_{i,j}$
- **The joint probability is** $P(X = x_i; Y = y_j) = \frac{n_{i,j}}{N}$



Marginal probability

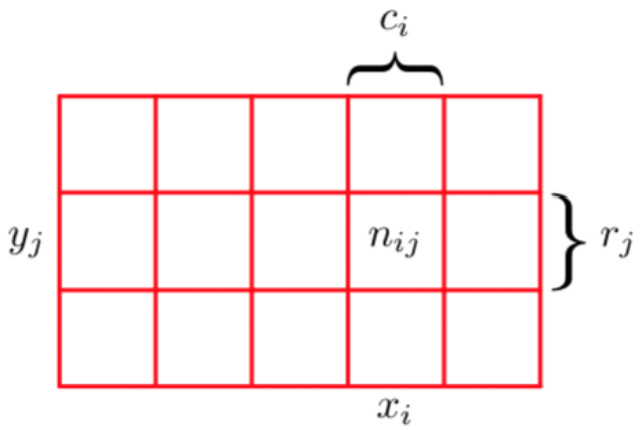
- The numer of time we find $X = x_i$ is c_i
- The numer of time we find $Y = y_j$ is r_j
- **The marginal probability is** $P(X = x_i) = \frac{c_i}{N} = \frac{1}{N}\sum_{j=1}^{L} n_{i,j} = \sum_{j=1}^{L} P(X = x_i; Y = y_j)$



Conditional probability

By considering only those instances for which $X = x_i$, the fraction of instances for which $Y = y_j$ is

$$P(Y = y_j \mid X = x_i) = \frac{n_{i,j}}{c_i}$$



Probability rules

Sum rule (marginal)

$$P(X) = \sum_{Y} P(X, Y)$$

Product rule

$$P(X, Y) = P(Y \mid X) P(X)$$
$$P(X, Y) = P(X \mid Y) P(Y)$$

Bayes' theorem

$$P(Y \mid X) = \frac{P(X \mid Y) P(Y)}{P(X)}$$

Conditional probability and Indipendence

Two random variables are independent when the probability of the one does not depend on the realization of the other

$$P(X \mid Y) = P(X)$$

so that conditioning does not add information.
This definition has the following consequence:

$$P(X, Y) = P(X \mid Y)P(Y)$$
$$\overset{indep.}{=} P(X)P(Y)$$

Distributions: Mass and Density

Discrete

probability mass function (p.m.f)

$$p_i := P(X = x_i) \text{ for } i = 1; \ldots, M$$

$$pi \geq 0 \text{ and } \sum_{i=1}^{M} p_i = 1.$$

Continuous

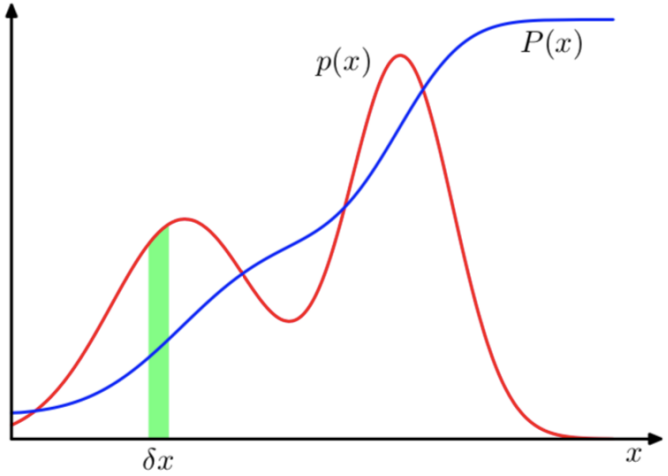
probability density function (p.d.f)

$$p(x)\delta x \text{ for } (x; x + \delta x)$$

and when $\delta x \to 0$ then $p(x)$ is the probability density over x.

$$p(x) \geq 0 \text{ and } \int_0^\infty p(x)dx = 1.$$

Distributions: PDF and CDF



The probability that x lies in the interval $(-\infty; z)$ is given by the cumulative distribution function (CDF) defined by

$$P(z) = \int_{-\infty}^{z} p(x)dxp(x)dx$$

which satisfies

$$dP(z) = dz\,|_{z=x} = p(x)$$

Expectation

The expectation is the weighted average value of some function of X, say f (X) under a distribution p(X) and is defined as:

Discrete $E[f] = \sum_x p(x) f(x)$

Continuous $E[f] = \int p(x) f(x) dx$

Suppose we are given N samples drawn from the p(X), then the expectation can be approximated as a finite sum over these points:

$$E[f] \approx \frac{1}{N} \sum_{n=1}^{N} f(x_n)$$

When $N \to \infty$ this converges to the true expectation, for both continuous and discrete variables.

Variance

The variance is a measure of variability of X around its expectation E[X] and it is defined as:

$$Var(X) = E[(X - E[X])^2]$$

This boils down to

$$Var(X) = E[X^2] - E[X]^2$$

Covariance

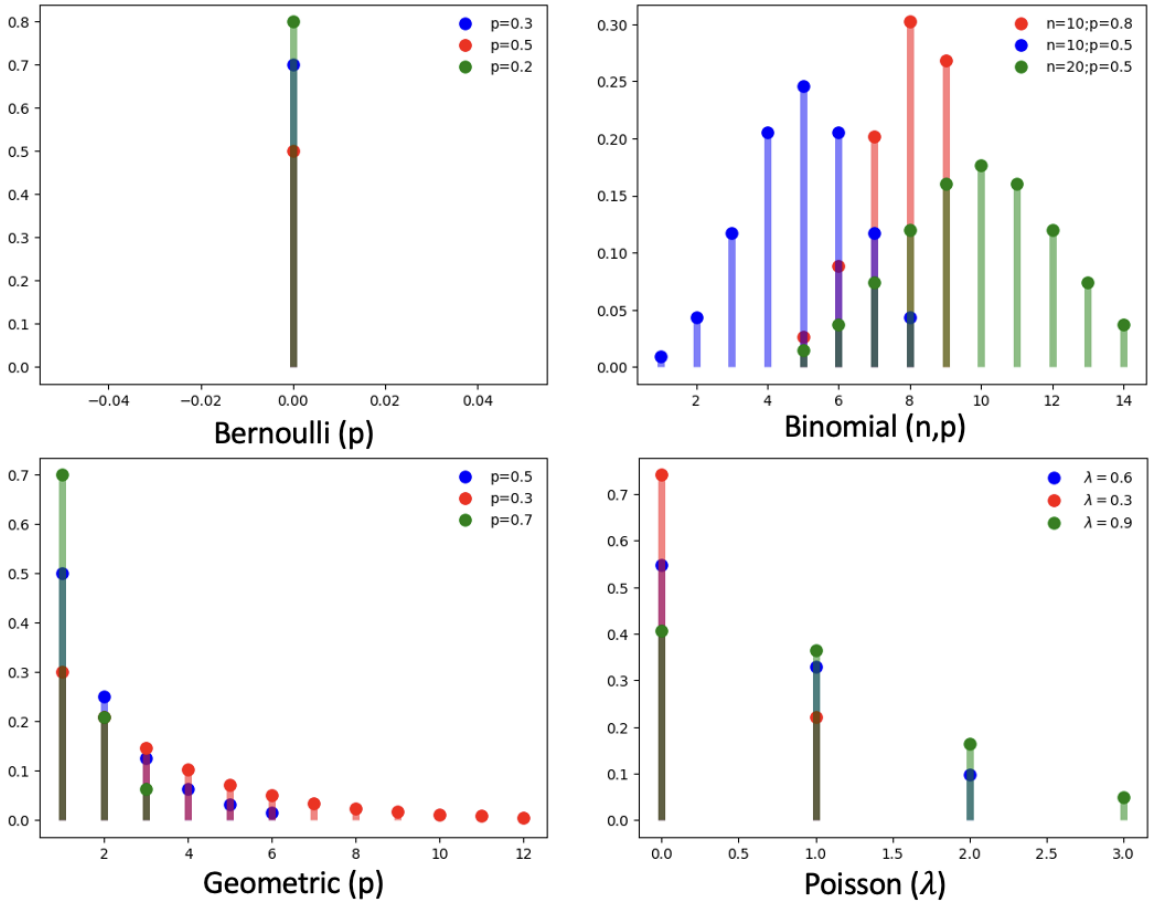
The covariance expresses the extent to which X and Y vary together and it is defined as:

$$Cov(X;Y) = E_{X,Y}[(X - E[X])(Y - E[Y])]$$
$$= E_{X,Y}[XY] - E[X]E[Y]$$

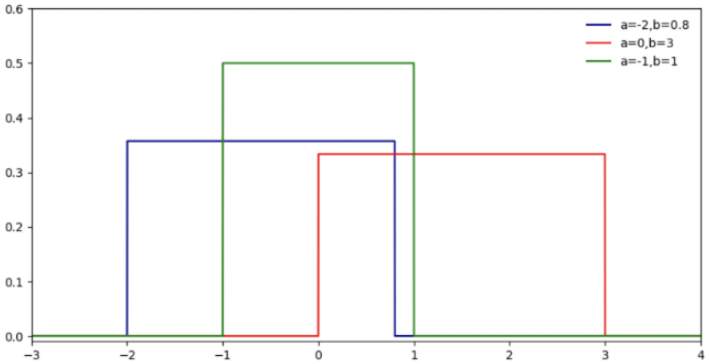
Note here that, when X,Y are independent random variables, P(X,Y) = P(X)P(Y), and hence

$$Cov(X,Y) = 0$$

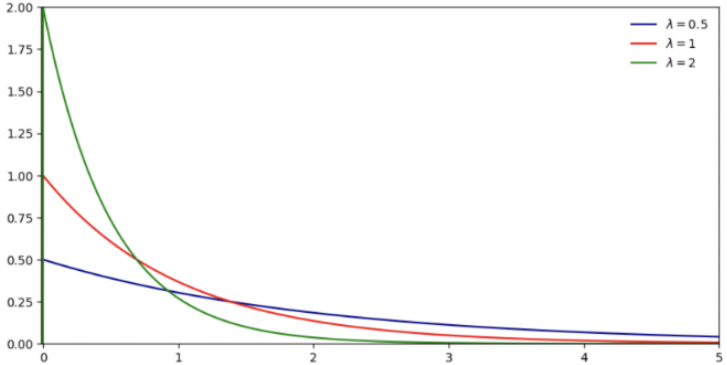
Some important distributions: discrete



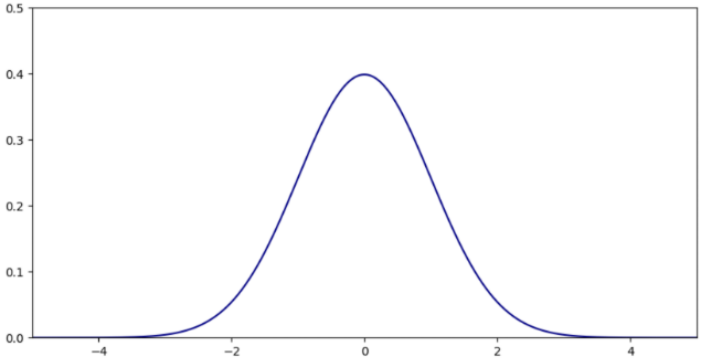
Some important distributions: continuous



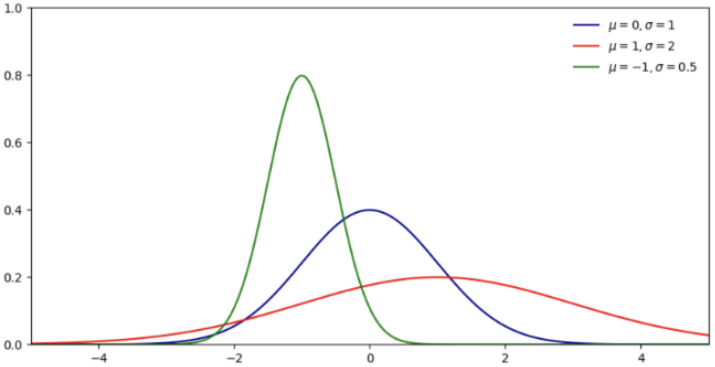
Uniform (a,b)



Exponential (λ)



Standard Normal (0,1)



Normal (μ, σ^2)

Limit theorems: LLN

LAW of LARGE NUMBERS (LNN): the average of a large number of i.i.d. random variables converges to the expected value.

The LNN has different formulations depending on the assumptions one decides to set. The most common assumption, as already mentioned, is to consider a succession of independent and identically distributed random variables.

Limit theorems: LLN

Weak Law of Large Numbers

Let $X1, X2, \ldots, Xn$ be i.i.d. random variables with a finite expected value $E[X_i] = \mu < \infty$.

For any $\epsilon > 0$,

$$\lim_{n \to \infty} P(|\overline{X} - \mu| \geq \epsilon) = 0$$

Limit theorems: CLT

CENTRAL LIMIT THEOREM (CLT): the sum of a large number of random variables has an approximately normal distribution (under some conditions).

Normalisation Suppose $X_1, \ldots, Xn$ are i.i.d. random variables with expected values $E[X_i] = \mu < \infty$ and variance $Var(X_i) = \sigma^2 < \infty$. Then:

- Sample mean has $E[\overline{X}] = \mu$ and $Var(\overline{X}) = \frac{\sigma^2}{n}$
- The normalised random variable $Z_n = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} = \frac{X_1 + \cdots + X_n - n\mu}{\sqrt{\sigma^2 n}}$ has mean $E[Z_n] = 0$ and variance $Var(Z_n) = 1$.

Limit theorems: CLT

Under the above assumptions (i.i.d. samples, finite mean, variance) the random variable $Z_n = \frac{\overline{X}-\mu}{\sigma/\sqrt{n}} = \frac{X_1+\cdots+X_n-n\mu}{\sqrt{\sigma^2 n}}$ converges in distribution to the standard normal random variable as n goes to infinity, that is

$$\lim_{n\to\infty} P(Z_n \leq x) = \Phi(x), \forall x \in \mathbb{R}$$

where $\Phi(x)$ is the standard normal CDF.

Exercices on probability

EX. 1

Consider a red box that contains 2 green apples and 6 oranges and a blue box that contains 3 green apples and 1 orange.

1. Pick the blue box, what is the probability of picking an orange?
2. Let us suppose now that we can pick one box or the other with the same probability. What is the probability of picking an orange?
3. Given that we have chosen an orange, what is the probability that the box chosen is the blue one?"

Exercices on CLT

EX. 1

A data-centre serves requests of service that arrive over time. Suppose the service time Xi for request i has mean $E[X_i] = 1$ [msec] and variance $Var[X_i] = 1$ [msec] (e.g. we can consider exponential service time). Let Y be the total time necessary to serve $n = 50$ requests. What is the probability that this time ranges between 40 and 60 [msec]?

EX. 2

data packet consists of $n = 1000$ bits. Due to noise, bit errors occur independently, with probability $p = 0,1$. What is the probability to have more that 120 errors within the same packet?